

MUHAMMAD TAHA NAWAB

Full-Stack Developer | AI/ML Engineer

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PROFESSIONAL SUMMARY

Computer Science student at COMSATS University Islamabad with hands-on experience architecting and shipping full-stack web applications using the MERN stack and production-grade AI/ML systems. Proficient in building scalable REST APIs, React SPAs, and machine learning pipelines — from data preprocessing and model training to live inference and deployment. Adept at integrating LLMs and Generative AI APIs into user-facing products. Seeking an internship or junior role to contribute to product development while deepening expertise in cloud-native MERN architecture and applied machine learning.

TECHNICAL SKILLS

Frontend: React.js, Vite, HTML5/CSS3, Bootstrap, JavaScript (ES6+), Responsive Design, Component Architecture

Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, JWT Authentication, WebSockets, BullMQ (async queues)

Databases: MongoDB, PostgreSQL, MySQL, Firebase/Cloud Firestore, TimescaleDB, pgvector, SQL Window Functions & Triggers, Schema Normalization, Query Optimization

AI / ML: scikit-learn, HuggingFace Transformers (BERT, RoBERTa, LayoutLMv3), OpenCV, MediaPipe, Tesseract OCR, Generative AI APIs (Gemini, OpenAI), Active Learning, Anomaly Detection, NumPy, Pandas

DevOps & Tools: Git, GitHub, Docker (basics), Linux/Bash, Postman, VS Code, npm/pip

CS Fundamentals: Data Structures & Algorithms, OOP, Computer Networks (TCP/IP, HTTP/HTTPS), Operating Systems, Agile/SDLC, Compiler Design

Languages: JavaScript, Python, SQL, Java, C++, Dart, Mojo

PROJECTS

MediPredict — AI Medical Diagnosis System

Jan 2025 – Present

[FastAPI](#) • [React](#) • [PostgreSQL](#) • [scikit-learn](#) • [JWT](#)

- Architected a full-stack disease-prediction platform processing **100+ symptom combinations** using a trained RandomForestClassifier, achieving **~90% classification accuracy** with confidence scoring on live inference.
- Engineered a FastAPI backend with JWT authentication, role-based access, and a normalized PostgreSQL schema (users, history, audit); REST endpoints consumed by a **responsive React SPA**
- Built an end-to-end ML pipeline — data preprocessing, feature engineering, model serialization — reducing average prediction latency to **under 200 ms** per request.

Customer Support Ticket System with SLA Monitoring

Jan 2026 – Present

[Node.js](#) • [Express.js](#) • [React](#) • [PostgreSQL](#) • [SQL Triggers](#) • [RBAC](#)

- Delivered an enterprise-grade ticketing platform capable of processing **10,000+ concurrent tickets** with automated SLA escalation — critical tickets auto-escalate to managers via PostgreSQL database trigger if unresponded within 4 hours.
- Implemented **role-based agent assignment** with row-level locking to eliminate race conditions on ticket claiming, plus a real-time dashboard surfacing open tickets, average response times, and per-agent workload via SQL window functions.
- Designed a **tamper-proof audit log** capturing every status transition (actor, timestamp, delta), enabling full compliance traceability across the ticket lifecycle.
- Optimized a normalized 6-table schema with indexed foreign keys, cutting average JOIN query time by **~40%** under load.

LearnWave — E-Learning Platform

Jun 2025 – Dec 2025

[React](#) • [Vite](#) • [FastAPI](#) • [PostgreSQL](#) • [JavaScript](#)

- Architected and deployed a full-stack instructor platform serving **course creation, enrollment, and content delivery** — React/Vite frontend + FastAPI backend connected to PostgreSQL.
- Implemented **modular component architecture** with ESLint-enforced code quality and responsive UI, reducing reported UI bugs by **60%** during QA.
- Built **CRUD course-management APIs** and a complete user authentication flow (register, login, JWT refresh) consumed by the React SPA.

ToxiGlow — AI-Powered Wound Analysis Platform

Apr 2026 – Jun 2026

[React 19](#) • [Vite](#) • [FastAPI](#) • [OpenCV](#) • [OpenAI GPT](#) • [SQLAlchemy](#) • [Python](#)

- Architected a **full-stack medical AI platform** for wound assessment — built for BioNova Innovathon 2026 — delivering clinical-grade tissue segmentation (granulation, slough, necrosis, epithelial) and infection risk detection in **under 5 seconds** per image.
- Engineered a **computer vision pipeline** (OpenCV + NumPy + Pillow) for automated wound measurement (area, perimeter, diameter) and a **severity scoring engine (0–100)** with annotated before/after visual output and downloadable PDF reports.
- Implemented **role-based access control** across 3 user tiers (Patient, Clinician, Admin) with an admin analytics dashboard serving live severity distributions, tissue pattern analysis, and CSV export; frontend bundle optimized to **289 KB (84 KB gzipped)**.

VisionAI Utility Application

Jan 2026

[Python](#) • [Tesseract OCR](#) • [TTS](#) • [Modular Architecture](#)

- Engineered a Python AI toolkit with **OCR (Tesseract) and Text-to-Speech** capabilities, processing **documents 3× faster** than manual transcription via automated pipeline.
- Separated concerns across dedicated modules (model handling, image preprocessing, custom UI), improving maintainability and enabling **plug-and-play extension** of new AI utilities.

Intelligent Warehouse Robot Simulation

Jun 2024 – Dec 2024

[Python](#) • [Tkinter](#) • [A*](#) • [BFS](#) • [Genetic Algorithm](#)

- Built an AI warehouse simulation with **autonomous robot navigation** using A* and BFS pathfinding, benchmarking Greedy Search, Hill Climbing, and Genetic Algorithms — A* delivered **optimal routes 95% of the time**.
- Developed an interactive **Tkinter GUI** enabling real-time visualisation of pathfinding decisions, reducing debugging cycles by **50%** compared to headless testing.

EDUCATION

BS Computer Science — COMSATS University Islamabad

Sep 2023

– Present

Relevant coursework: Data Structures & Algorithms, Database Systems, Software Engineering, Computer Networks, Operating Systems, Compiler Design, AI Fundamentals

Intermediate — FSC / ICS — Punjab Group of Colleges

Sep

2021 – Jul 2023

CERTIFICATIONS

- **AI Fluency for Students** — Anthropic (Oct 2025)
- **CSS, Bootstrap, JavaScript & Python Stack** — Udemy (Oct 2025)
- **Implementing MVC ToDo App with Flask** — CodeSignal (Aug 2025)
- **Git Version Control for Beginners** — CodeSignal (Aug 2025)
- **Computer Science Introduction** — Udemy (Nov 2025)

LANGUAGES

English (Professional) • **Urdu** (Native) • **Punjabi** (Native)